

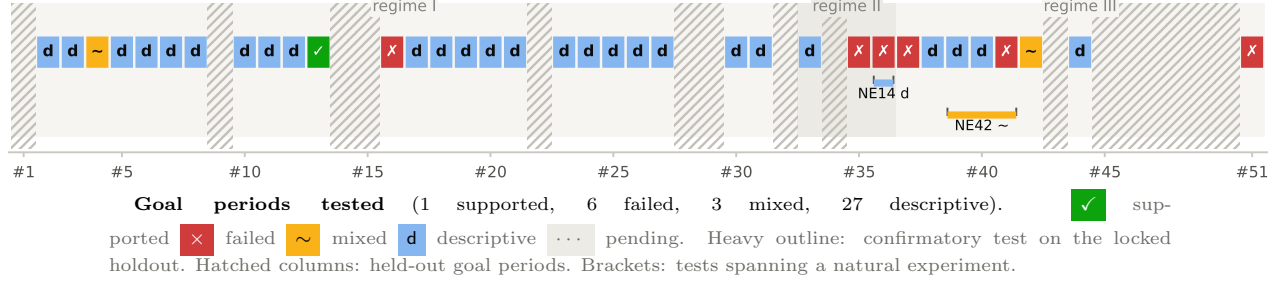
H99: Fluctuations predict relaxation (mean-field Glauber)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 65% Faithfulness 3/5 Usefulness 2/5

Bottom line. The regime-III talk memory that mean-field Glauber misses is coupling delayed by one read-out call. It survives scheduler removal, respects the room and address gates, and matches H67's read-out gain. The tenfold nudge excess was an alignment artefact.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Mean-field Glauber dynamics ties the gain g read from equal-time fluctuations to how much slower the collective mode relaxes than one agent. Round 1 (70 units) found that in regime III the collective talk mode keeps more lag-1 memory than this predicts ($\Delta\rho_1$, post hoc). The open rival was a slow common drive, the scheduler. Round 2 re-indexed talk on each agent's call clock, removed the scheduler field, and added partition contrasts. **The excess is read-out coupling, not the scheduler.** It survives every removal (+0.052 median; #51 +0.042 [0.014, 0.070]), lives inside rooms ($\Pi \approx 15$), and equals what H67's read-out gain produces ($g_{1,\text{fit}}/g_{\text{lag}} = 0.96$). The $\times 9.5$ nudge result is withdrawn. The holdout is not run.

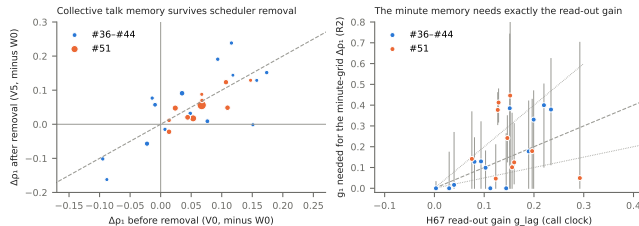
PHYSICAL MODEL

Models 02, 01 and 09. Talk spin $s_i(t)$ per minute on the call-based all-present window, or Y_{ic} per receiving call c . Read-out coupling acts at the recipient's next call: messages posted before t_c are read; messages posted during the call are in flight. A slow common drive acts on every agent at once, whether or not it reads. Each unit's null is the W0 world: independent talk at its real calls.

$$\Delta\rho_1 = \rho_c(1) - \rho_\perp(1)^{1-g_X}, \quad \Delta\rho_1^c = \Delta\rho_1 - \langle\Delta\rho_1\rangle_{W0}$$

$$P(Y_{ic}=1) = a_{i,d,\kappa} + \sum_h \beta_h R_{ic}^h + \beta_0 P_{ic}^0 + \beta_X X_{ic}^1 + \rho Y_{i,c-1}$$

$$g_1 = \bar{m} \bar{r} (\beta_1 - \beta_X), \quad \Omega = \sum_k G(k) / [G(p) \sum_k S(k) / S(p)]$$



Observables. Left: one point per regime-III unit (size $\propto$ trimmed calls). The W0-corrected collective memory before (V0) and after (V5) removing the call-window, day-edge, exogenous and activity-field structure. Points stay near the diagonal, so the removals change little. Right: the hop-1 gain that a call-skeleton simulator needs to reproduce each unit's minute memory, against H67's read-out gain. Bars are CIs; dotted lines mark $\times 2$.

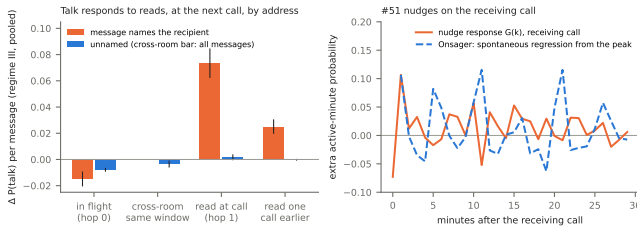
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For a monitor: in regime III, read the collective talk memory as read-out coupling of size $g \approx 0.11\text{--}0.14$ (push amplified $\times 1.15$), not as a scheduler artefact. Report $\Delta\rho_1^c$ next to g_{lag} ; they agree. For an operator: naming the recipient is the lever. A named message raises the reader's talk chance by 7.3 points at its next call and by 2.5 points one call later; an unnamed one by 0.2. A nudge's response is about what the target's own spontaneous wake predicts; it is not tenfold.

RESULTS VS PREDICTIONS

FAITHFULNESS SCORECARD

Prediction (dated)	Observed	Verdict
R1 synthetic (real skeletons)	J_1^* only in coupling worlds; g_1 recovered +10%; room partition fails only under a room-local drive	3/5 pass, 2 marginal
Q-S1 survives removal (V5)	median +0.052; +0.042 [0.014, 0.070]	#51 supported
Q-S2 no tracking of c_x	Spearman -0.10 (p 0.64)	supported
Q-P1 room gate (minute grid)	same 0.0165, cross 0.0011 [-0.003, 0.005]	supported
Q-P2 address gate (call clock)	named $\times 10$; placebos negative	mixed
Q-R1a self memory > 0	-0.048 (refractory)	failed
Q-R1b not instantaneous	in flight -0.015 vs read $+0.073$ (named)	supported
Q-R1c kernel $> \text{one call}$	named 0.025 (ratio 0.34); unnamed 0	mixed
Q-R2 $g_{1,\text{fit}} \approx g_{\text{lag}}$	ratio 0.96 (median, 24 units)	supported
Q-R4a nudge $\Omega \leq 2$	1.29; A_{30} 0.44 [-0.41, 1.23] (post hoc)	supported
Q-R4b human $\Omega \in [0.5, 2]$	talk per call 2.53 [1.63, 3.44] (post hoc)	mixed



Second figure. Left: regime-III pooled talk response per message by window on the call clock. Messages a call cannot read (in flight, other rooms) carry a small negative association. Read messages carry a positive one, almost all from messages that name the recipient, which keep acting one call later. Right: #51 nudge activity response aligned on the target's receiving call, with the matched spontaneous regression (Onsager) scaled to its peak.

GOAL PERIODS IN DETAIL

G02 d descriptive
G03 d descriptive
G04 $\sim$ mixed
G05 d descriptive
G06 d descriptive
G07 d descriptive
G08 d descriptive
G10 d descriptive
G11 d descriptive
G12 d descriptive

+27 more periods: see the card.

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

$\Delta\rho_1$ was chosen after round-1 data (A2). The round-2 tests were pre-registered, with estimator fixes found in the synthetic (B1). The room partition cannot exclude a room-local drive; the call-clock gates do. The in-flight placebo is negative in regime III, so the in-flight-gated gain (0.23) overstates; the room-gated one (0.11) is used. R4 normalization changed after a first run (B2, post hoc), and the nudge activity CI is wide. Regime I stays untestable on this design.

ROUND 2

Holdout on #43 (V5 $\Delta\rho_1^c$, room partition, $g_{1,\text{fit}}/g_{\text{lag}}$) via a re-frozen confirm script. Fit the full per-call kernel of named messages against H40's decay. Identify the negative in-flight field.

- **Essence.** a dynamical check that tells a coupling from a field without inferring J, on the clock at which agents actually update.
- **Round-3 redirects (newest).**
- **H99-R6** Holdout. Run V5 $\Delta\rho_1^c$, the room partition and $g_{1,\text{fit}}/g_{\text{lag}}$ on #43 (regime III) through a re-frozen confirm.py, after Vivian's sign-off.
- **H99-R7** Named hop-2 kernel. Fit the full per-call kernel of named messages (hops 1–5) and test it against H40's per-call decay ϕ .
- **H99-R8** Negative in-flight field. Every unreadable message around a call lowers its talk probability, in every room (β -0.005 to -0.008).
- **H99-R3** (regime-I power) and H99-R5 (content) remain open.